

Neck Node Levels & Head and Neck Cancer Lymphatic Drainage Patterns

Integrated Biomedical and Clinical Sciences: OMFS 513

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Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

1. Discuss the anatomy of the oral cavity and the oropharynx.
2. Review premalignant and malignant lesions of the oropharyngeal region.
3. Review lymphatic drainage of the head and neck region.

Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1



**Head and neck Cancer Awareness Month is
recognized in April.**

***Help Make Cures Possible for ALL
Cancers***

Head and neck cancer

Symptoms and common locations

Nasal cavity

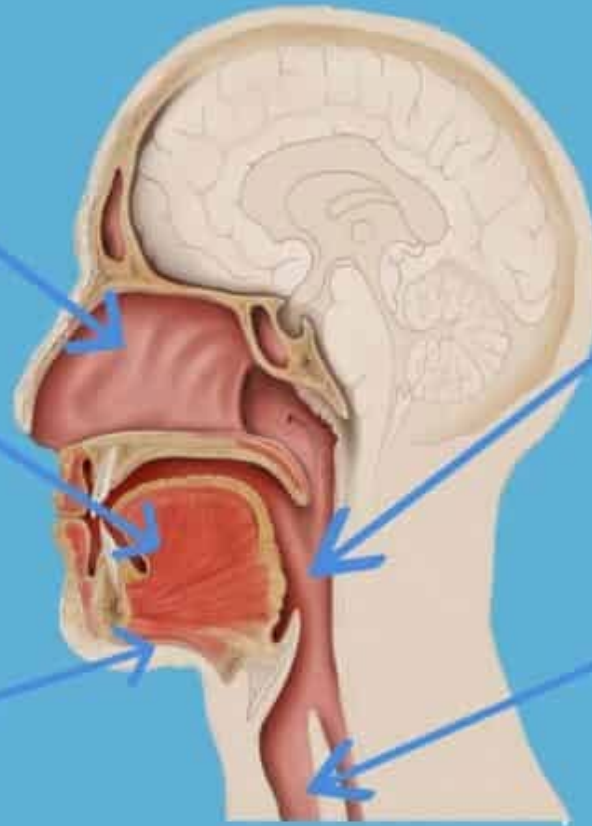
- Persistent blocked sinuses
- Untreatable sinus infections
- Headaches

Oral cavity

- Swellings on the jaw or lining of the mouth
- Bleeding
- Red patches on the gums

Salivary gland

- Swelling around the jawbone
- Persistent numbness or paralysis of the muscles in the face, chin or neck



Pharynx

- Difficulty breathing or speaking
- Persistent pain in the neck or throat
- Headaches
- Hearing problems

Larynx

- Causes pain in ears and throat when swallowing

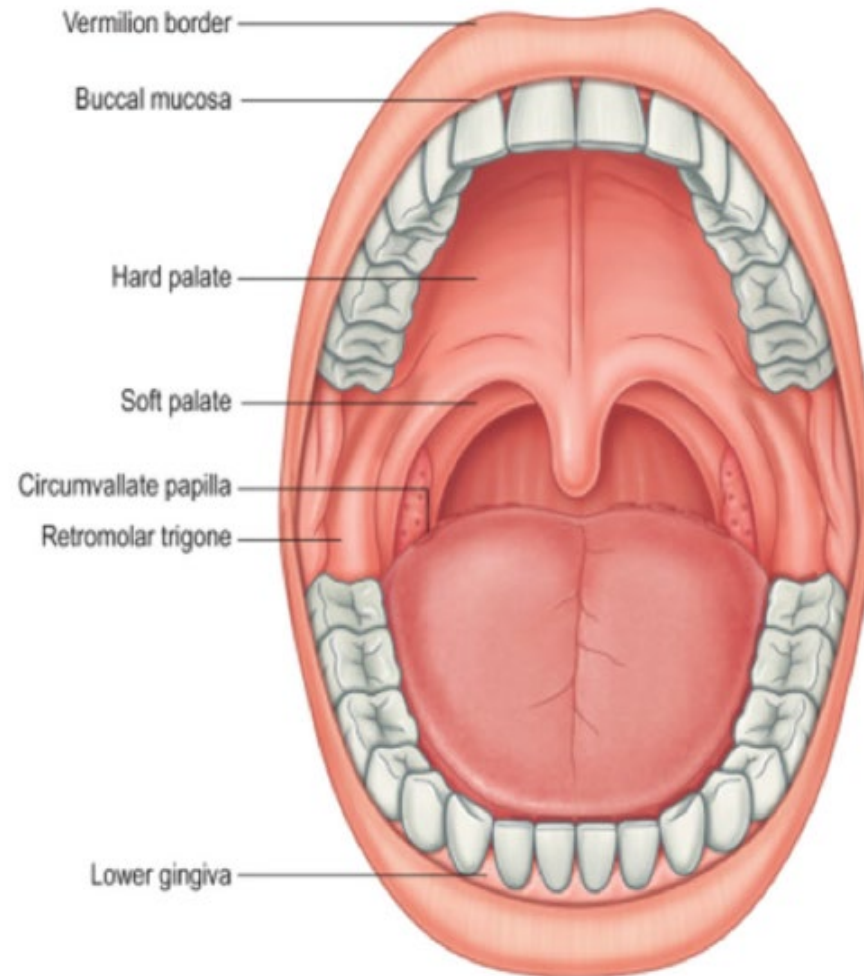
Figure 2. Adapted from "Head and neck cancer infographic". Oncology Central. <https://www.oncology-central.com/infographics/head-and-neck-cancer-infographic/>
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Head and Neck Cancer Statistics

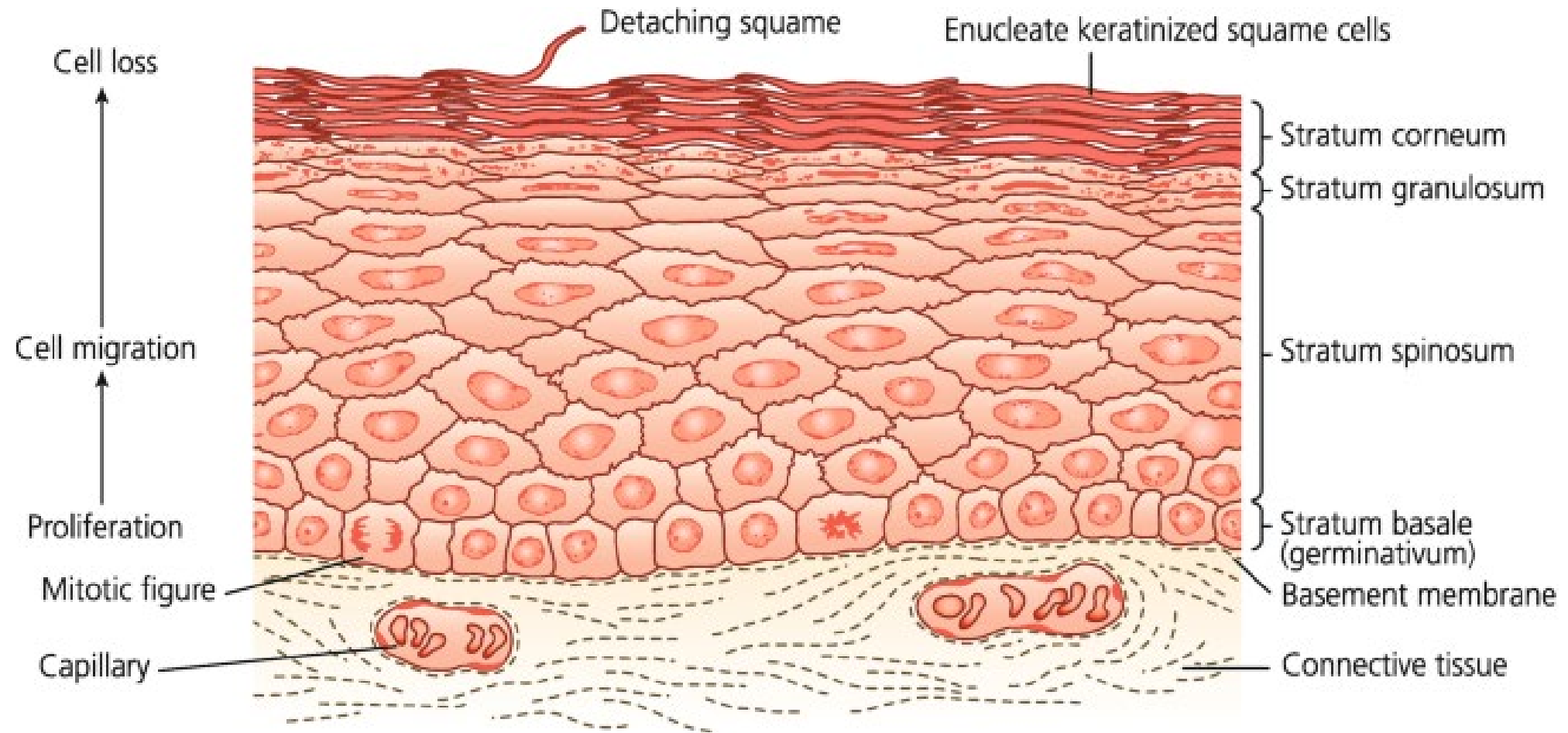
Cancer Subtype	Annual Incidence			Annual Mortality		
	Male	Female	Ratio	Male	Female	Ratio
Lip and Oral Cavity	264,211	113,502	2.3 : 1	125,022	52,735	2.4 : 1
Larynx	160,265	24,350	6.6 : 1	85,351	14,489	5.9 : 1
Nasopharynx	96,371	36,983	2.6 : 1	58,094	21,914	2.7 : 1
Oropharynx	79,045	19,367	4.1 : 1	39,590	8553	4.6 : 1
Hypopharynx	70,254	14,000	5.0 : 1	32,303	6296	5.1 : 1

[Cancer Types | Head and Neck Cancer - National Foundation for Cancer Research](#)

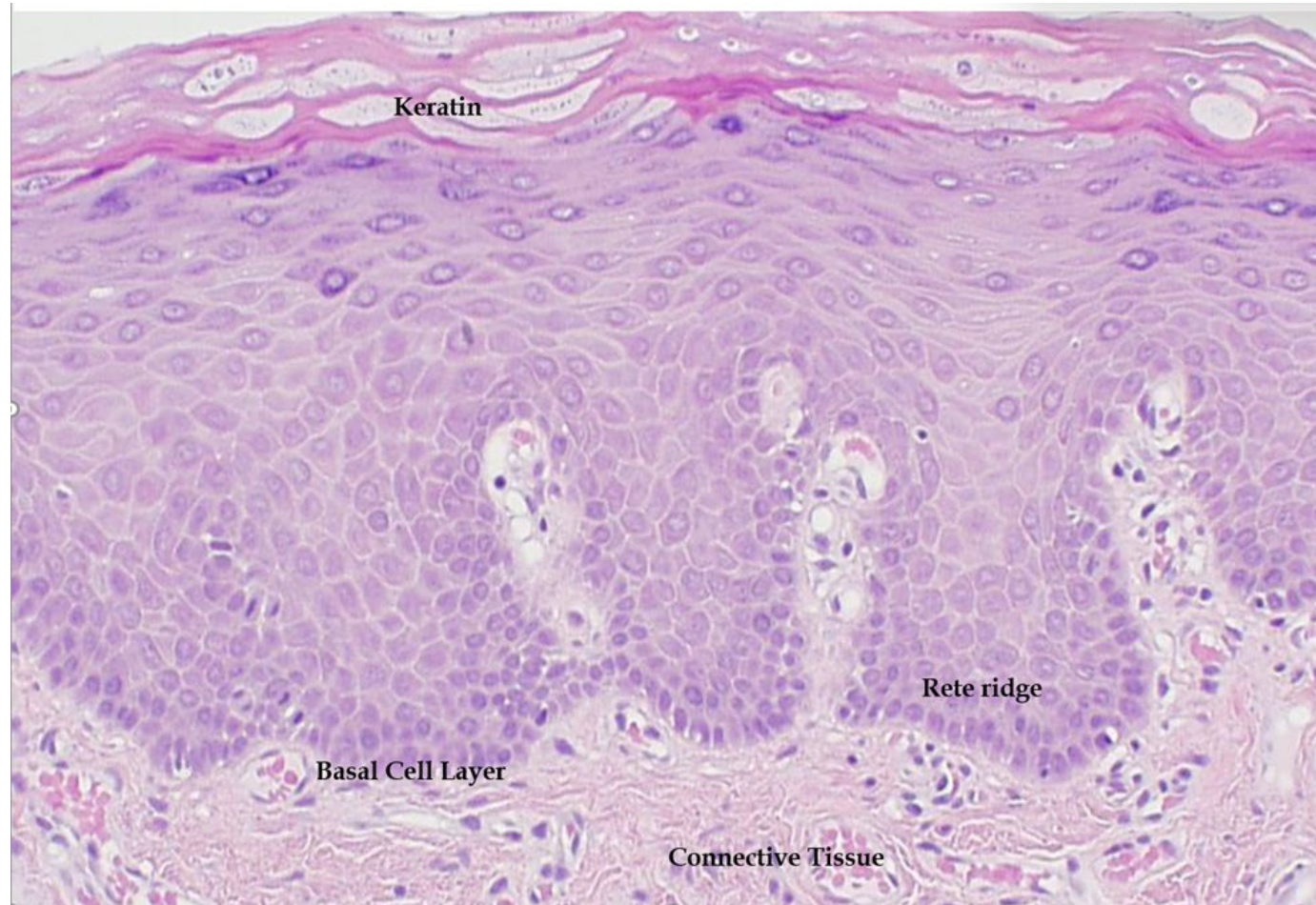
Oral Cavity Proper



Oral Mucosa



Oral Mucosa Histology



Oropharynx

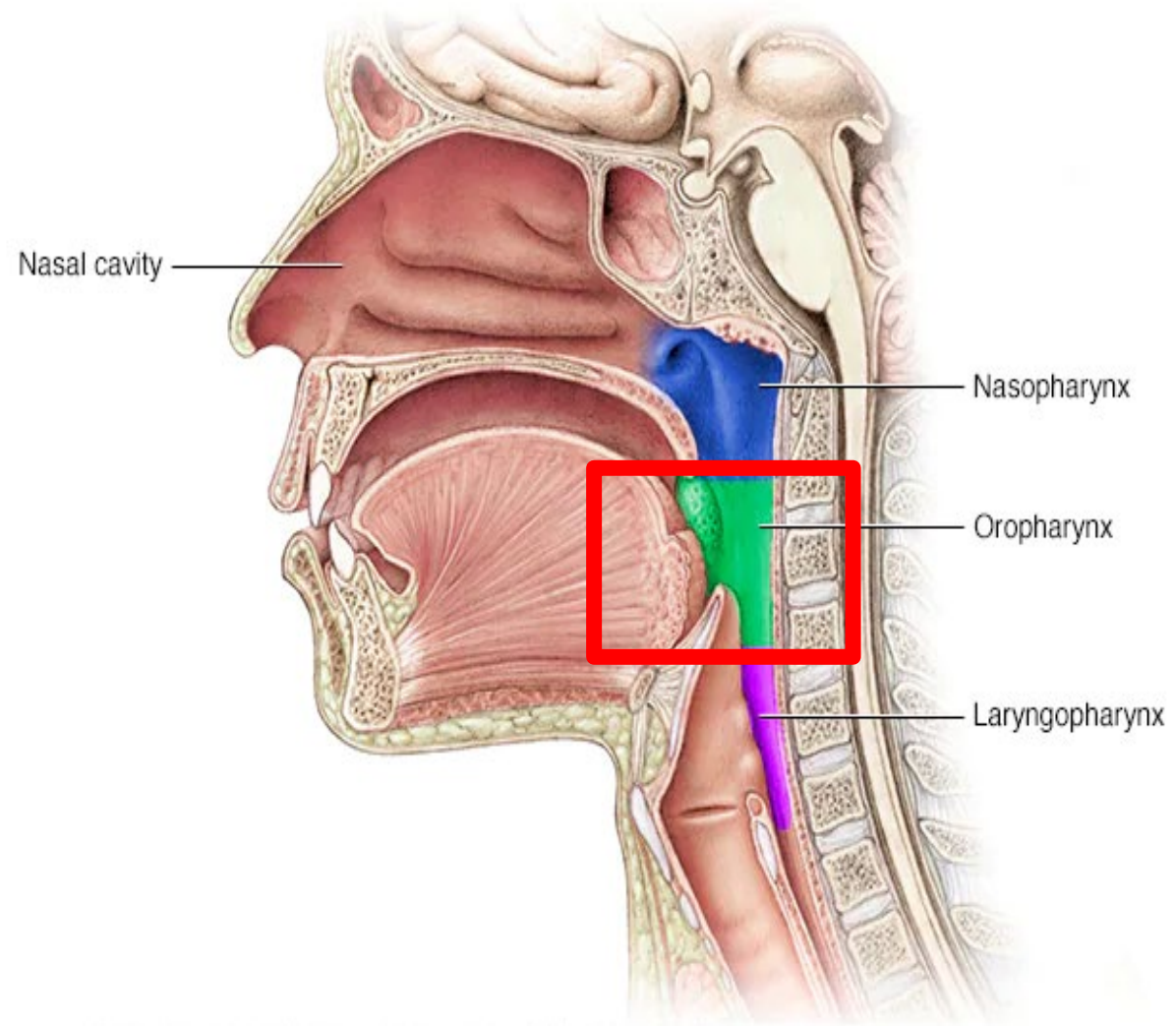
Oropharynx consist of the:

- Posterior 1/3rd of the tongue
- Tonsils
- Posterior pharyngeal wall

HPV-Related Squamous Cell Carcinoma
(HPV+ SCC)

Non-HPV-Related Squamous Cell
Carcinoma (HPV- SCC)

- Human Papilloma Virus-16



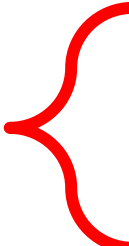
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Risk factors

- Tobacco use
- Alcohol use
- Exposure to radiation and chemicals
- Vitamin/nutritional deficiencies
- HPV Infection
- Compromised immune system

Oral Potentially Malignant Disorders

Table 1: Oral Potentially Malignant Disorders



Erythroplakia

Erythroleukoplakia

Leukoplakia

Oral submucous fibrosis

Dyskeratosis congenita

Smokeless tobacco keratosis

Palatal lesions (associated with reverse smoking)

Chronic candidiasis

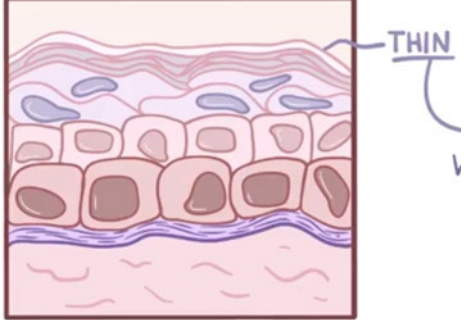
Lichen planus

Discoid lupus erythematosus

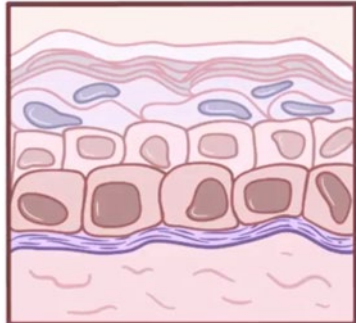
V. Thorburn et al. *Oral Health*. Oral Cancer and the Race Against Time: Use of Psoriasin as a Predictive Marker to Enhance Assessment of Cancer Risk in Oral Lesions

Leukoplakia

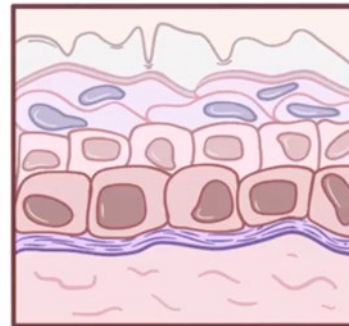
- Clinical term for a premalignant white patch that cannot be classified as any other disease.



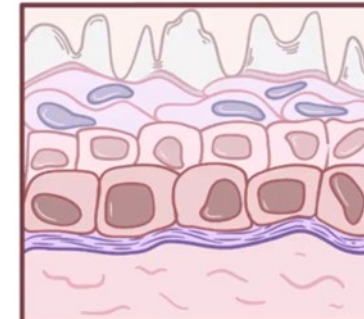
THICK LEUKOPLAKIA



NODULAR LEUKOPLAKIA

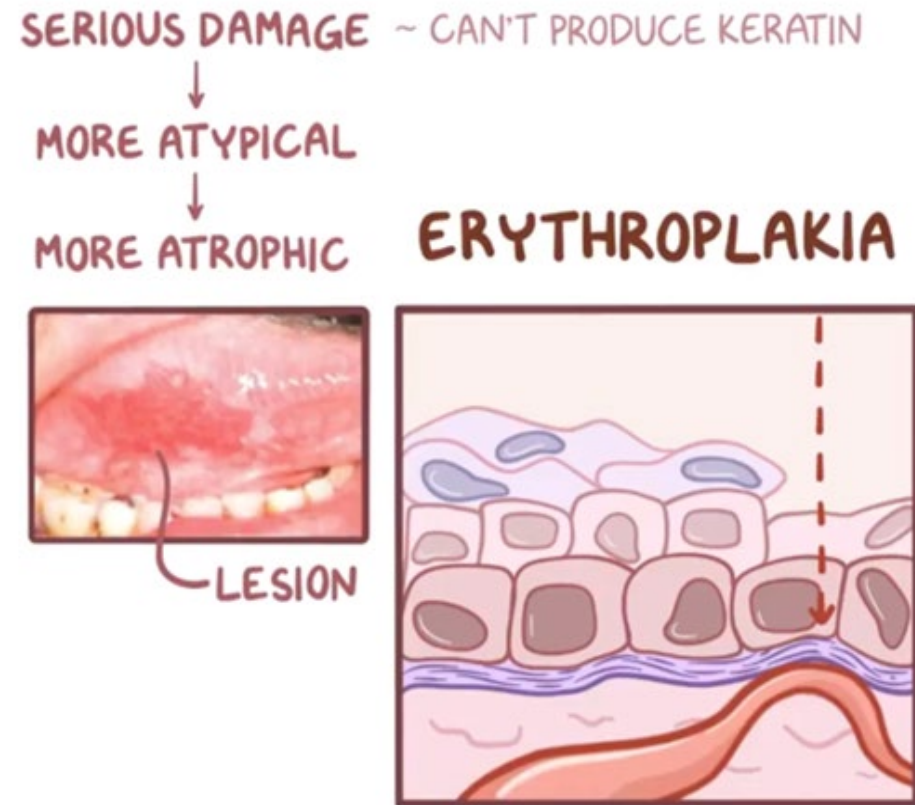


VERRUCOUS LEUKOPLAKIA



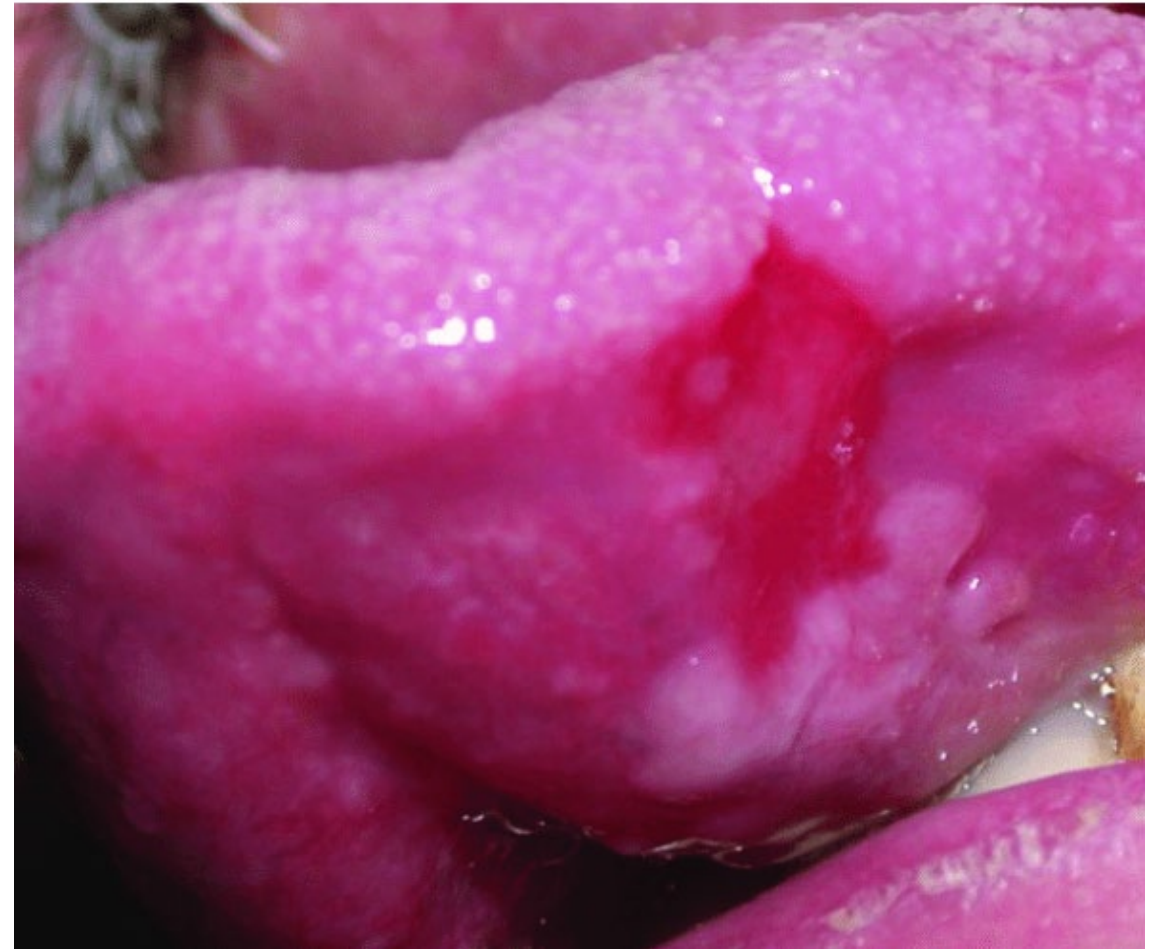
Erythroplakia

- Red, nonspecific patch or plaque which cannot be classified clinically and pathologically under any other disease. It is considered a premalignant lesion.
- Not as common as leukoplakia
- Higher association with squamous cell carcinoma
- Erythroleukoplakia-mixed red and white lesion



Erythroleukoplakia

- Clinical term for mixed red and white lesions.

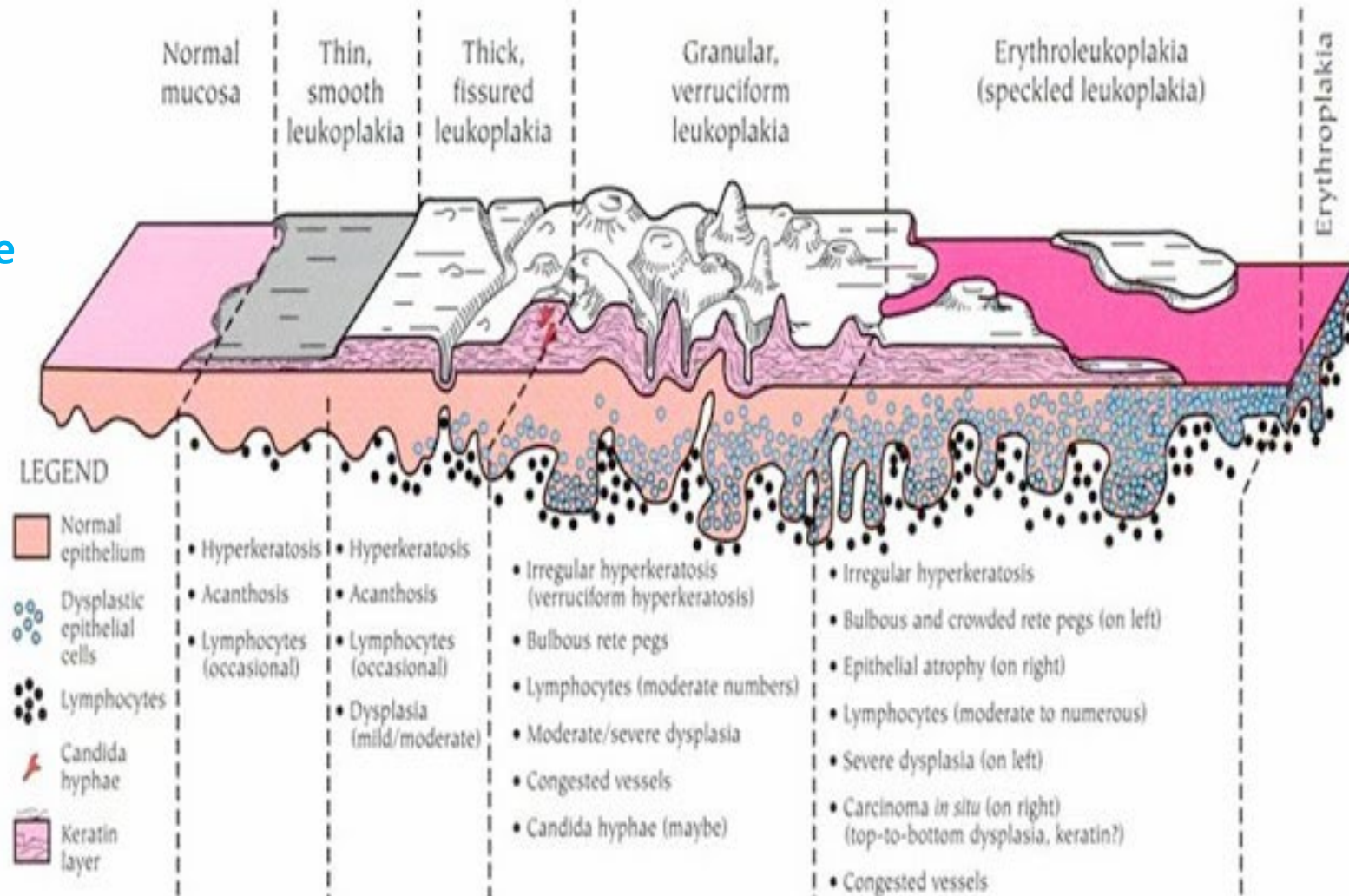


Perks et. al Oral Cancer Prevention. (2019)

Clinical/Histologic changes from normal to premalignancy

Clinical Appearance

Histology

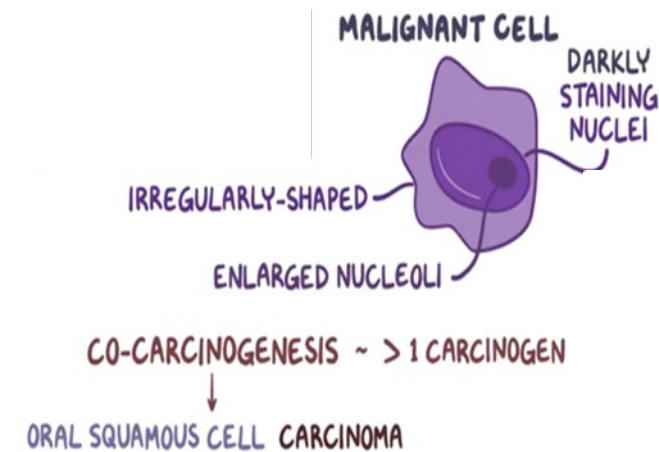


Neville et al. Oral and Maxillofacial Pathology. 3rd Edition Fig. 10-69.

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Dysplasia

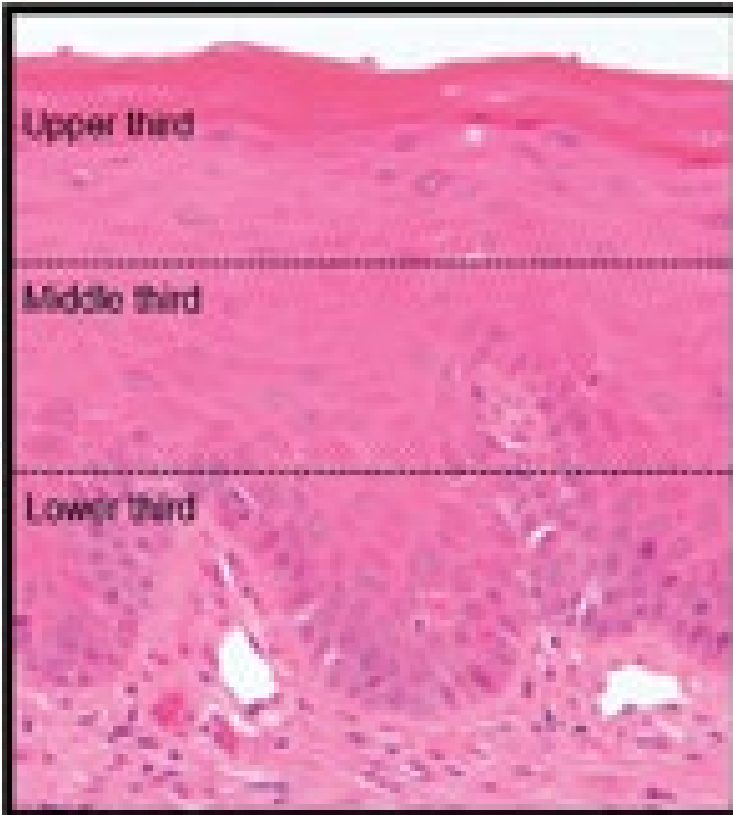
- Abnormal cellular growth and development
- Delay in maturation and differentiation of epithelial cells
- Levels of dysplasia
 - Mild
 - Moderate
 - Severe



Osmosis.org

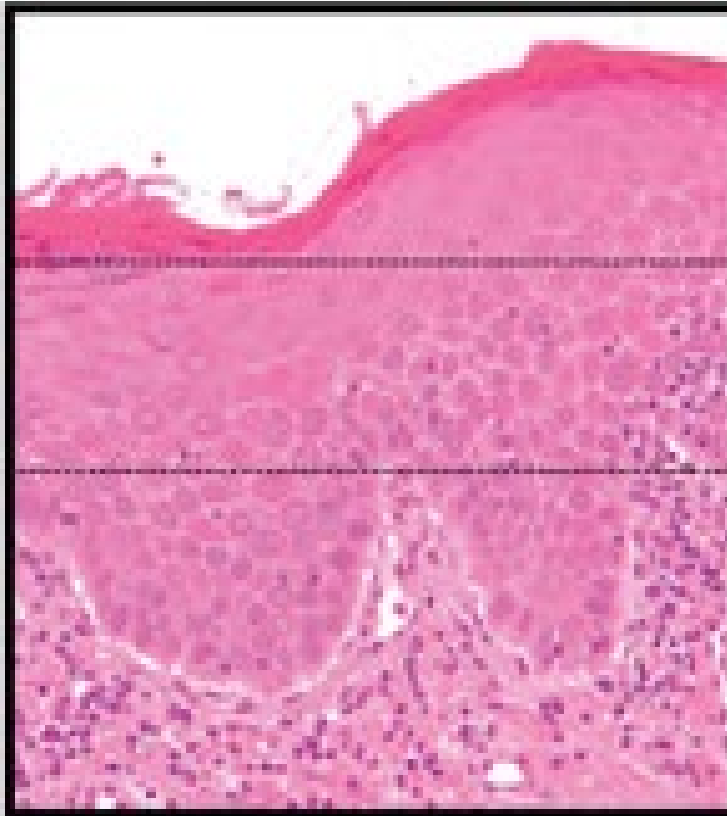
Levels of Dysplasia

Mild Dysplasia



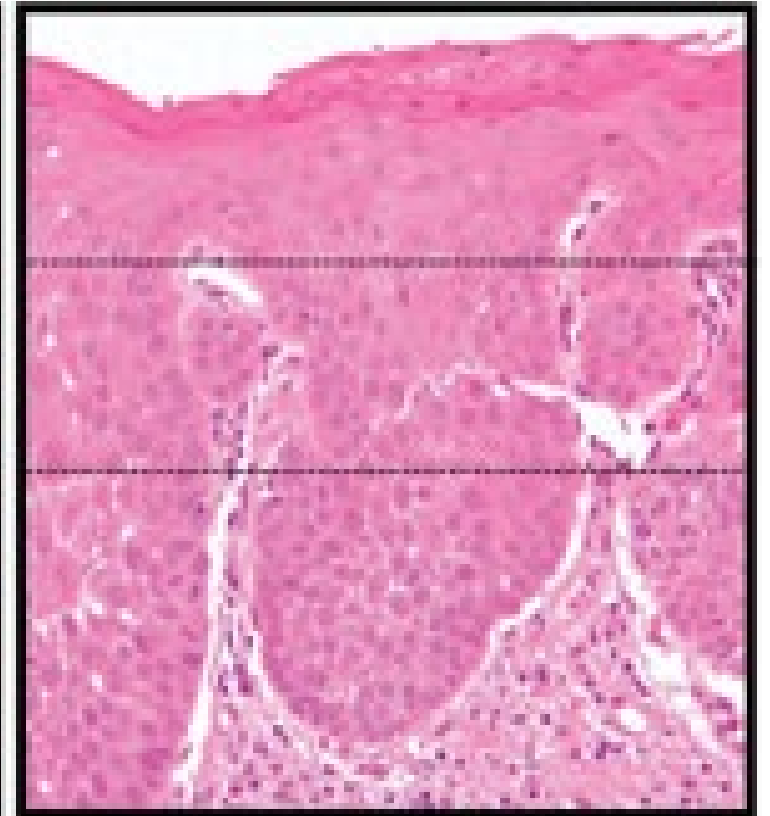
Mild- the lower **1/3rd** of the lower portion of the epithelium shows abnormal changes.

Moderate Dysplasia



Moderate- the lower **2/3rd** of the epithelium exhibits dysplastic changes.

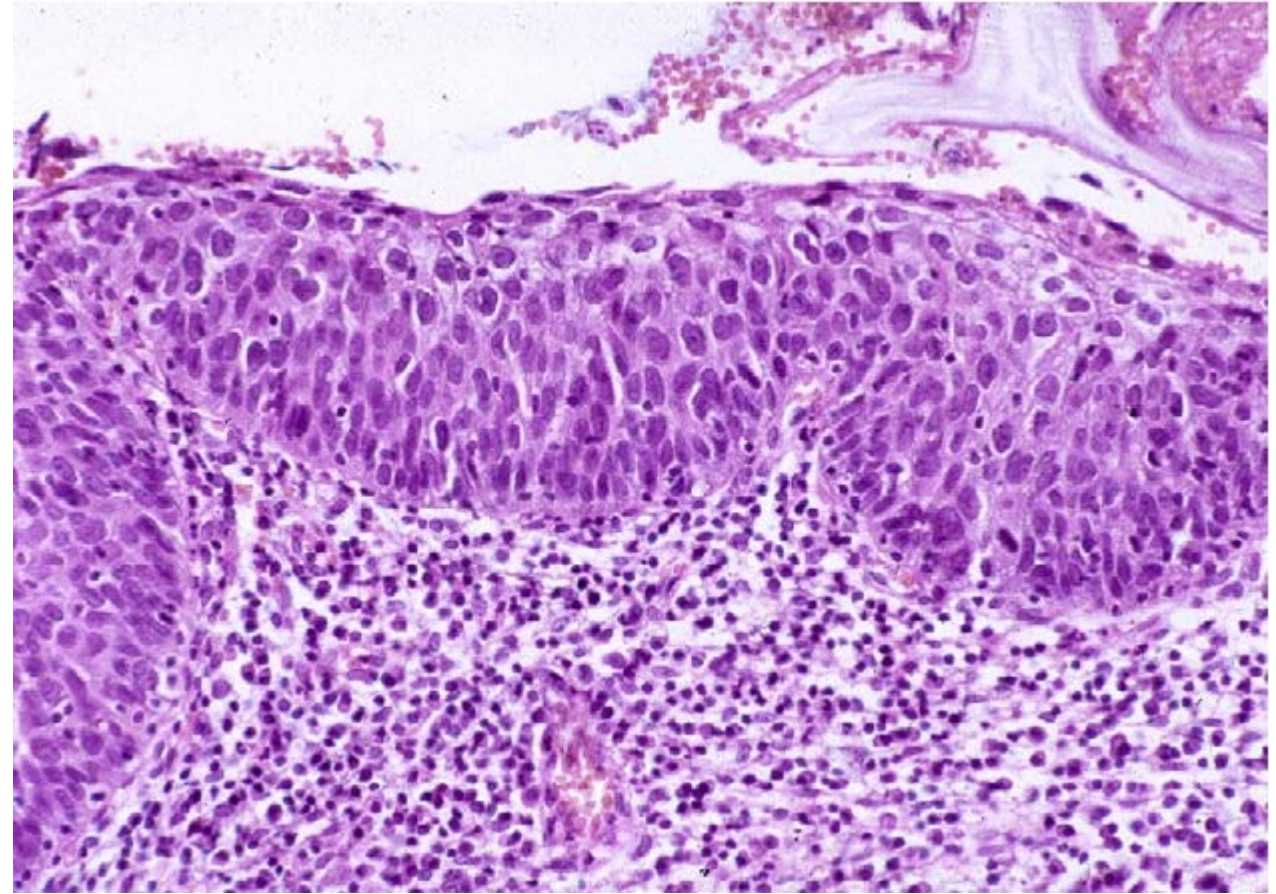
Severe Dysplasia



Severe- **Most but not the entire** epithelium has dysplastic changes.

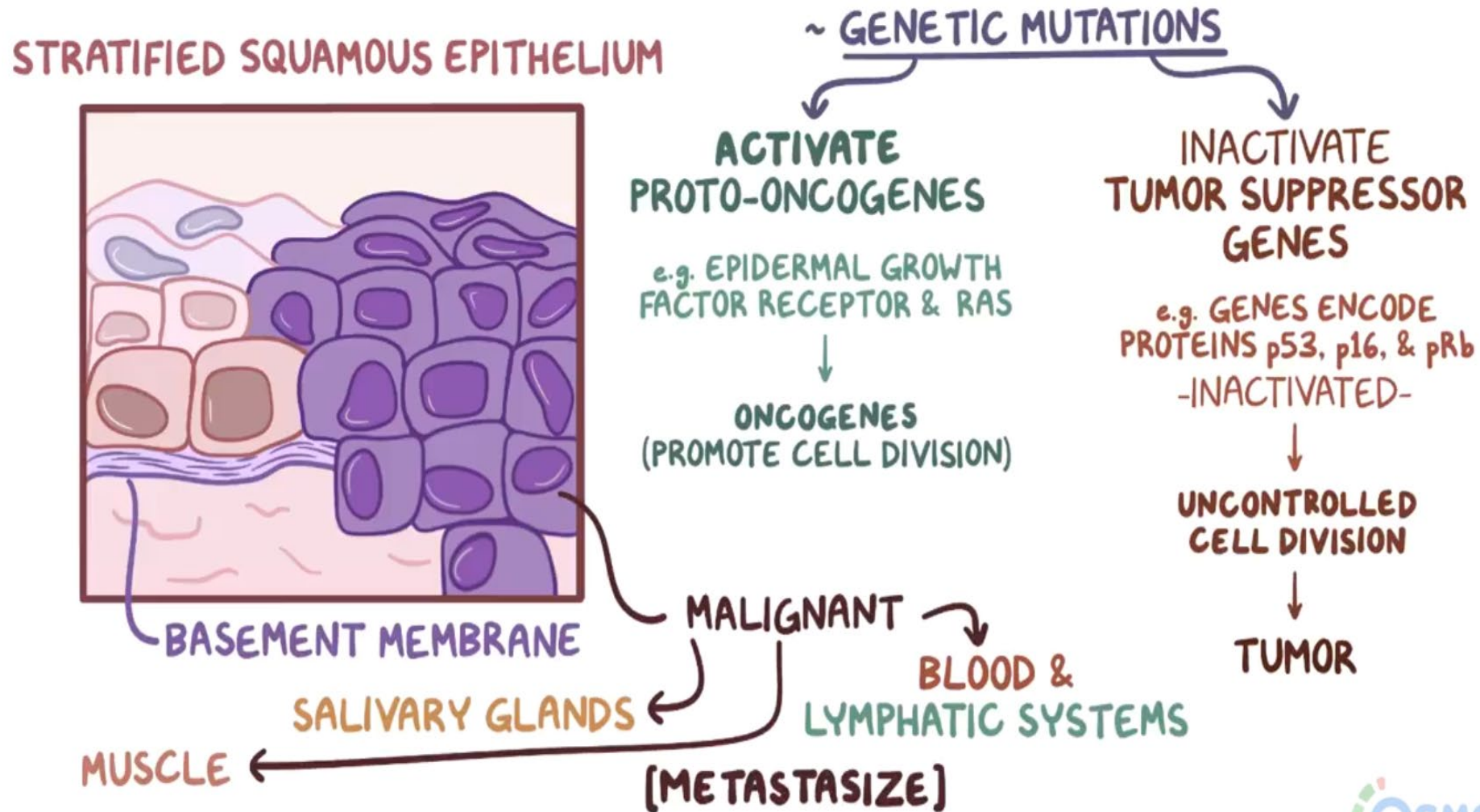
Carcinoma-in-situ

Altered cells involve the entire epithelium, but there is **no invasion** into the underlying connective tissue.



Bouquot, Jerry E et al. "Epithelial dysplasia of the oral mucosa—Diagnostic problems and prognostic features." *Current Diagnostic Pathology* 12 (2006): 11-21.

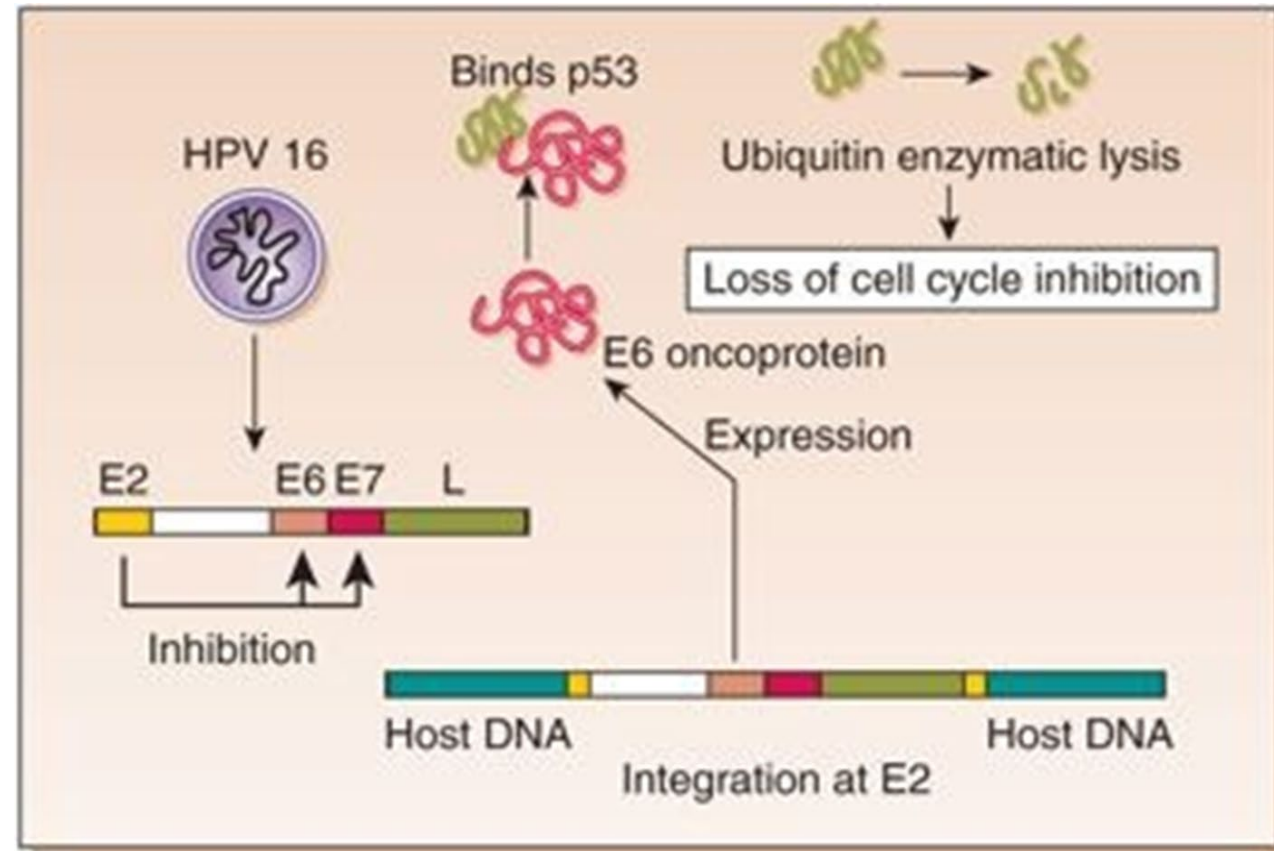
MAJORITY: SQUAMOUS CELL CANCERS



HPV and Squamous Cell Carcinoma

HPV infection of the oral epithelial cells occurs in a stepwise fashion:

1. The viral particles infect the basal cells through an abrasion or wound.
2. The viral genome integrates into the host genome.
3. The viral E6 and E7 genes from HPV encode for oncoproteins which inhibit the action of the host tumor suppressor genes p53 and retinoblastoma.
4. Uncontrolled cell division and tumor formation.



Demographics

Cancer Site	Oral Cavity	Oropharynx	
		HPV(-)	HPV(+)
Demographics ⁵	• Smoker/drinker • Older • More African-Americans • Lower SES • Lower education	• Smoker/drinker • Older • More African-Americans • Lower SES • Lower education	• Nonsmoker • Male • Younger • Caucasian • Multiple partners ← • Higher SES • Higher education
Common Locations ⁶	• Oral Tongue	• Pharyngeal wall • Soft Palate	• Tonsil • Base of tongue
Common Presentations ⁶	• Soreness with red or white spots	• Sore throat • Dysphagia • Otalgia	• Painless neck mass

headandneck.org

Squamous Cell Carcinoma

SYMPTOMS

- ~ NUMBNESS or CHANGES in SENSATION
- ~ HOARSE VOICE
- ~ PAIN or DIFFICULTY with CHEWING / SWALLOWING
- ~ PAINLESS & PAINFUL LUMPS, SORES, or DISCOLORATIONS

SIGNS of PRECANCEROUS DEVELOPMENT

- * LEUKOPLAKIA
- * ERYTHROPLAKIA
- * ERYTHROLEUKOPLAKIA
- * ABNORMAL TISSUE or MASSES

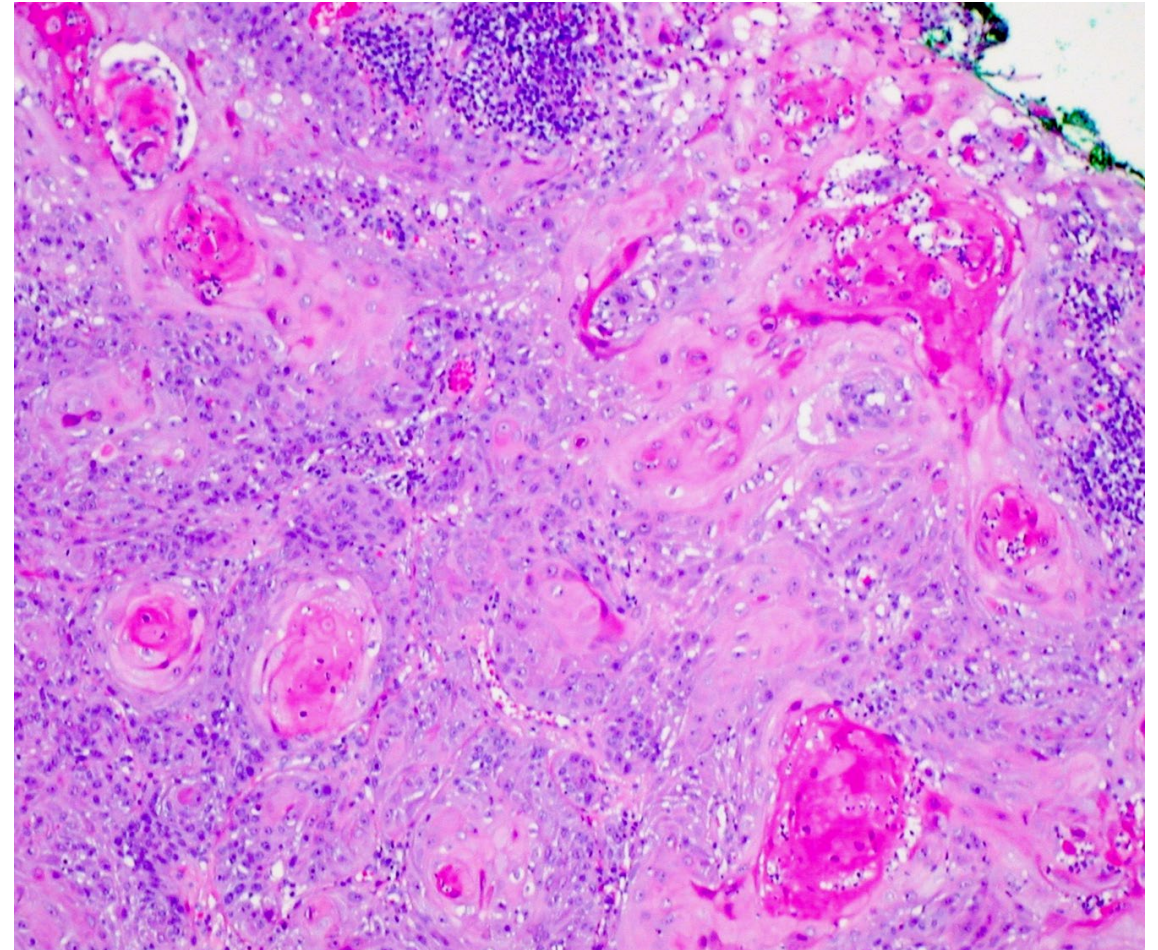
Conventional Oral Squamous Cell Carcinoma

- Most common malignancy of the oral cavity
- Male predilection
- Occurs in adults 40 years of age and older
- Most common sites
 - Lateral and ventral border of the tongue
 - Floor of the month
 - Lower lip (vermilion border)

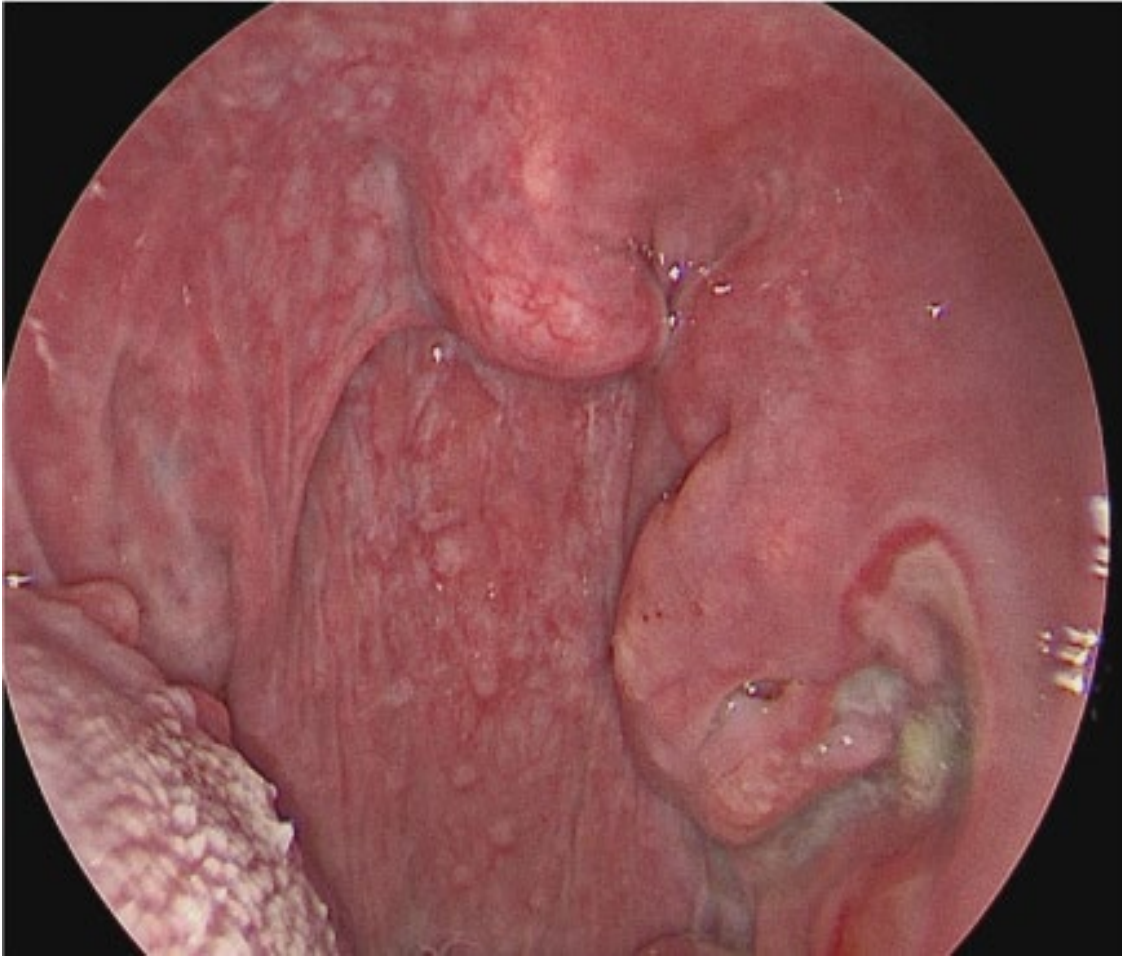


Squamous Cell Carcinoma

- Histology
 - Epithelial islands in connective tissue
 - Pleomorphic cells, hyperchromatic nuclei
 - Atypical mitotic figures and keratin pearl formation
 - Vascular and/or perineural invasion

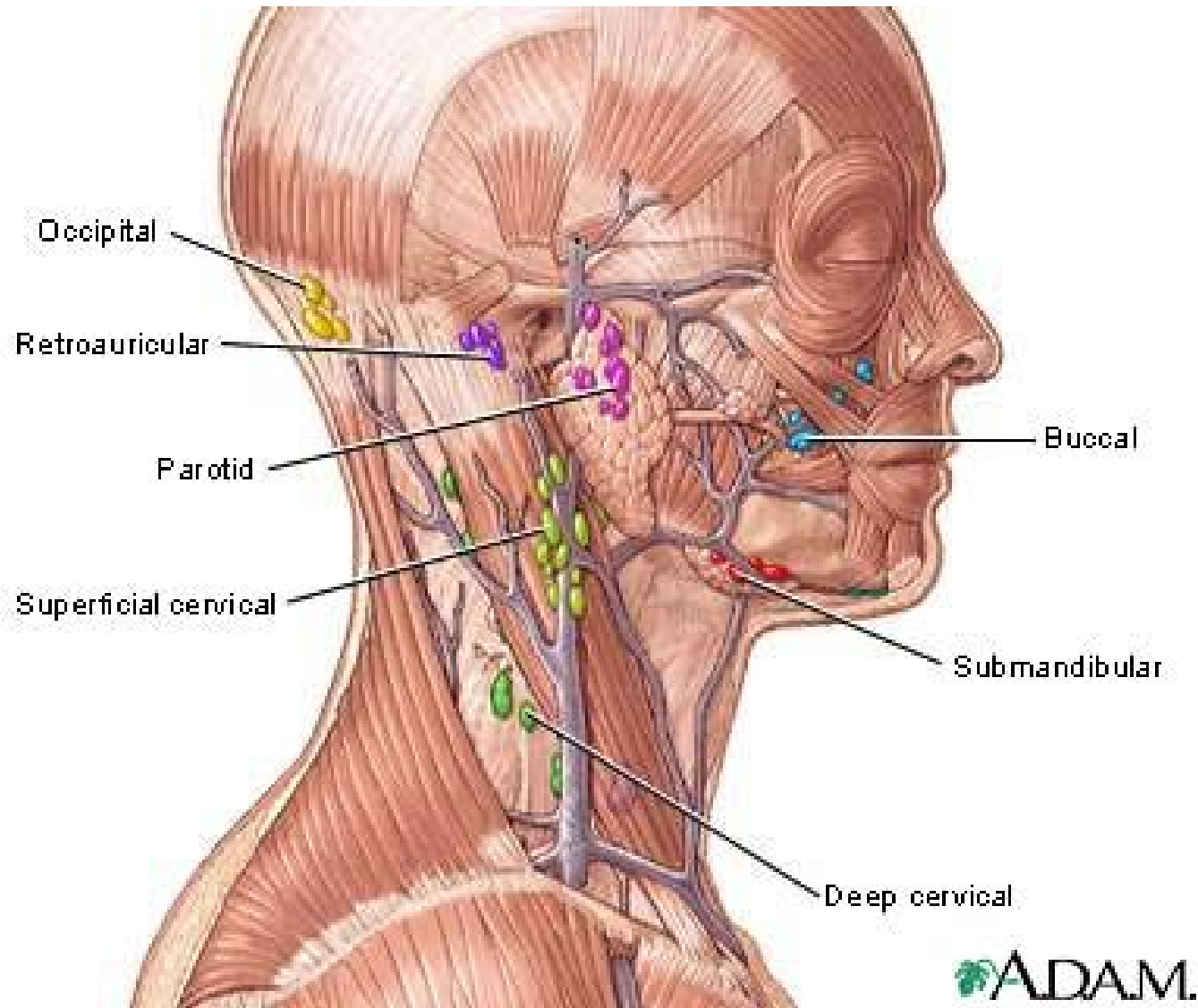


HPV-Related Squamous Cell Carcinoma (HPV)



Courtesy of Dr. Adepitan Owosho

Lymph nodes



SUPERFICIAL RING OF NODES

The superficial ring surrounds the transition area of neck to head and is arranged into five main groups (*Figure 8-5*; see *Table 8-3*). These nodes are palpable when infected.

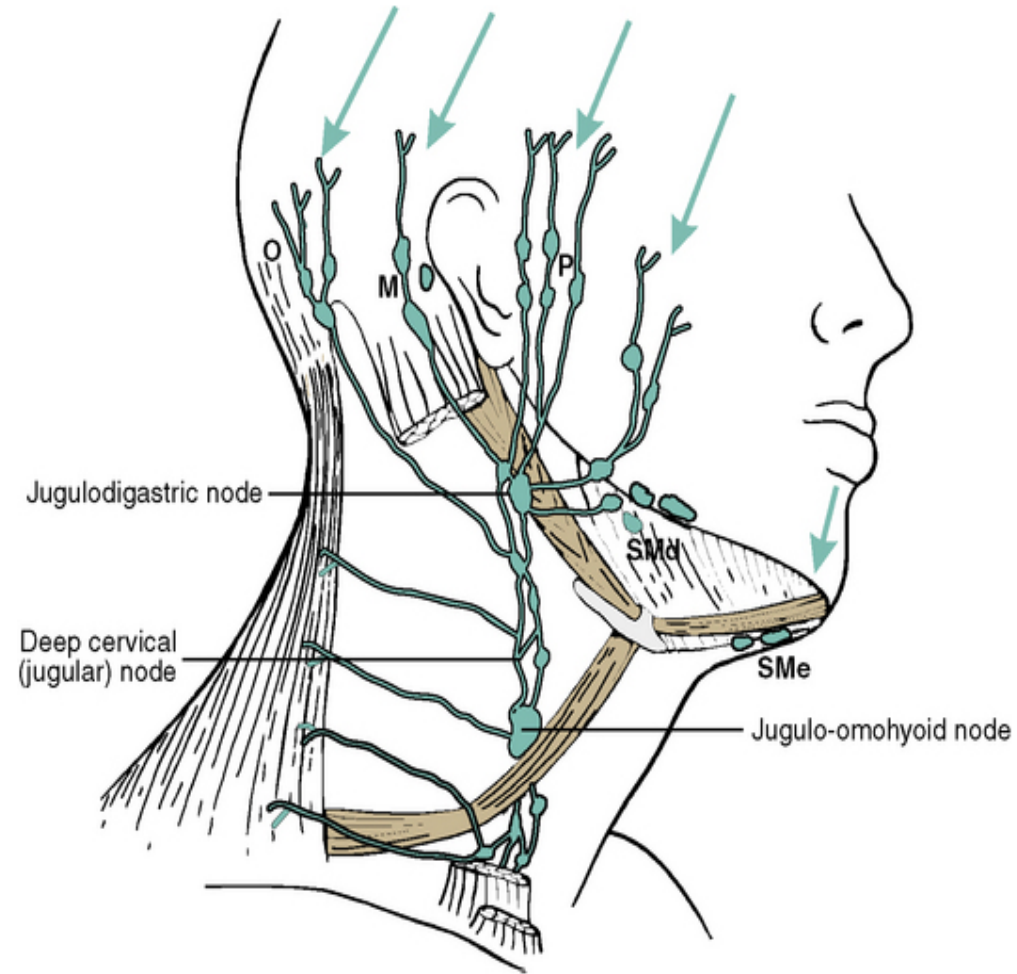
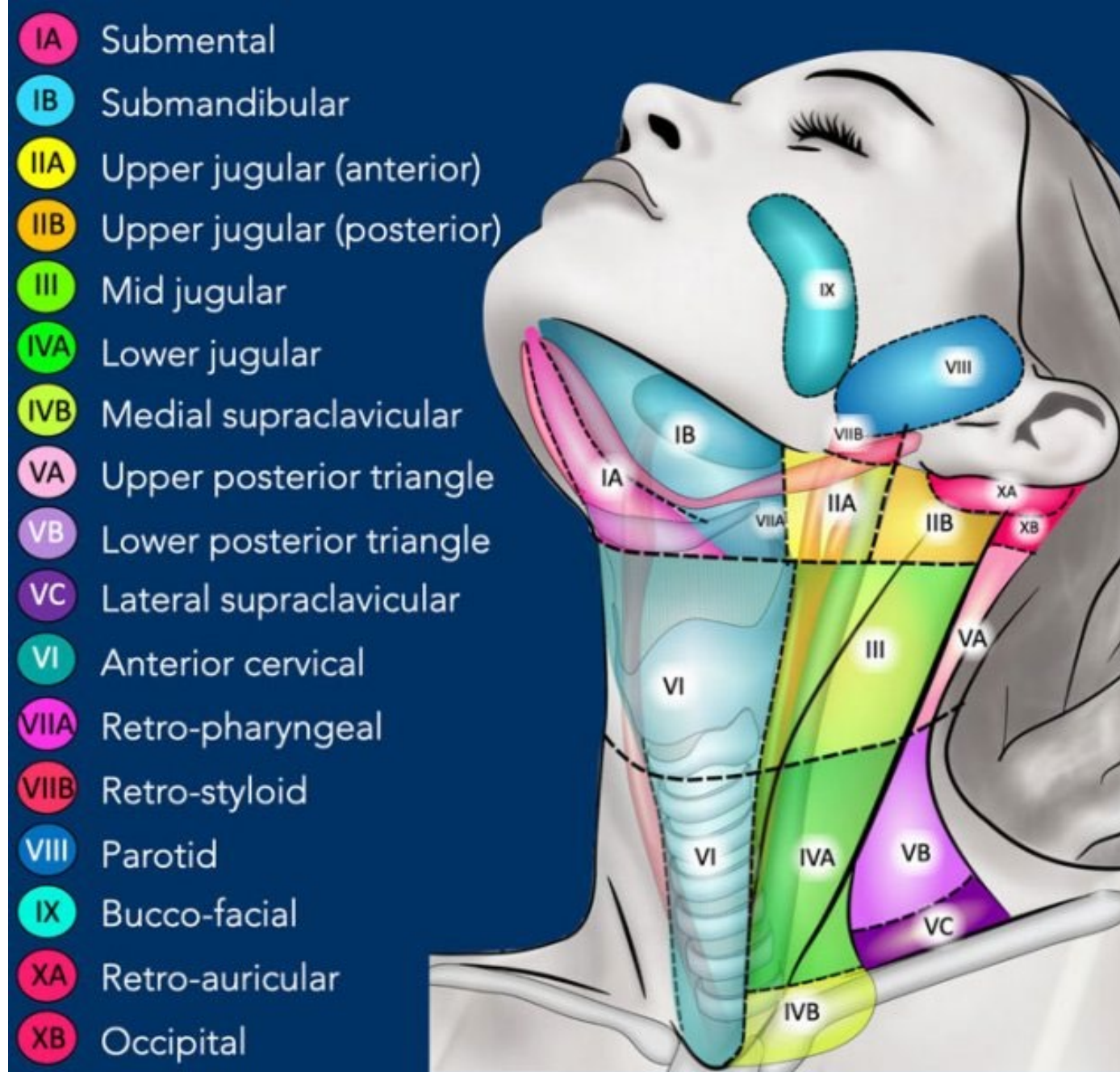


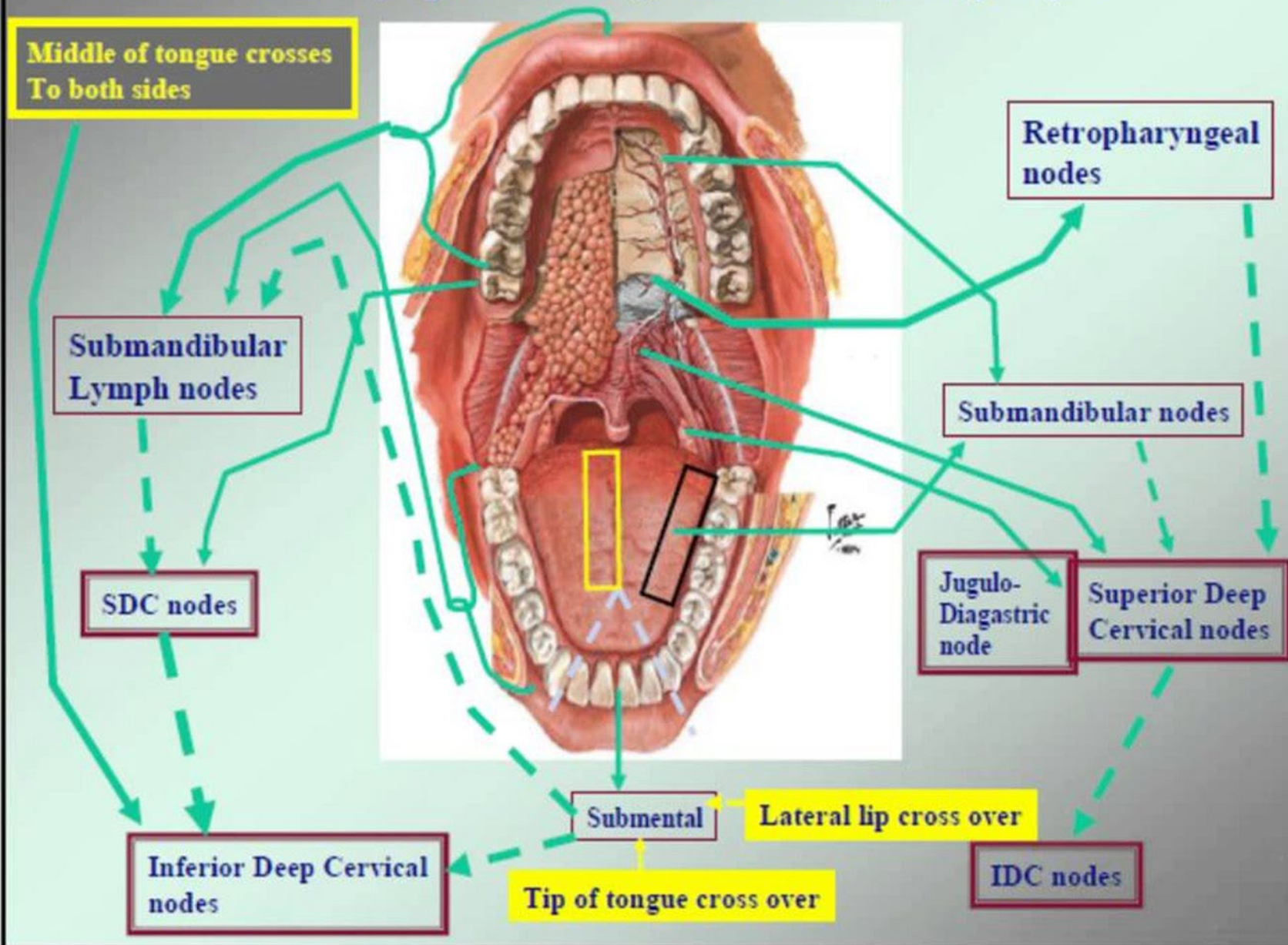
Figure 8-5 Lymphatic drainage of the face. SMe, Submental nodes; SMd, submandibular nodes; P, parotid (preauricular) nodes; M, mastoid (postauricular) nodes; O, occipital nodes.



Group of Lymph Nodes	Receives Drainage of Primary Malignancies from these Locations
Level IA: Submental Level IB Submandibular	IA: Floor of mouth, anterior tongue, mandibular alveolar ridge, and lower lip. IB: Submental nodes and facial nodes (buccal mucosa, nasal mucosa, eyelids, and conjunctiva), upper lip, and marginal areas of the lower lip
Level II (A&B): Deep Cervical-Upper Jugular	Oral cavity, nasopharynx, oropharynx, hypopharynx, larynx, and major salivary glands
Level III: Deep Cervical-Middle Internal Jugular	Oral cavity, oropharynx, nasopharynx, larynx, and hypopharynx
Level IV (A&B): Deep Cervical- Lower Internal Jugular	Hypopharynx, larynx, thyroid, cervical esophagus, trachea, and rarely the anterior oral cavity
Level V (A,B,&C): Posterior Triangle	Nasopharynx, paranasal sinuses, oropharynx, and thyroid, and posterior scalp
Level VIa: Anterior (Central) Compartment Level VIb	VIa: Lower lip, soft tissue of chin and advanced carcinoma form the hard and soft tissue of the mandible VIb: lower lip floor of mouth, and anterior tongue, thyroid, parathyroid, cervical esophagus, and larynx

Group of Lymph Nodes	Receives Drainage of Primary Malignancies from these Locations
Level VII (A&B): Superior Mediastinal Nodes	Nasopharynx, pharyngeal wall, oropharynx (tonsils and soft palate)
Level VIII: Parotid Group	Orbit, external auditory canal, and parotid gland
Level IX: Buccofacial Group	Facial skin, nose, buccal mucosa, maxillary sinus lesions invading the soft tissue of the cheek
Level X: Posterior Skull Group A: Retroauricular (Mastoid) and B: Occipital	Xa: Retro-auricular skin Xb: Occipital skin

General schema for lymphatic drainage of Oral Cavity and pharynx



TMN- Staging

T-Tumor size and extension into anatomical structure

N- Lymph node involvement

M-Metastasis

- M₀ – no metastasis
- M₁ – metastasis present

Change	7th Ed. (2010)	8th Ed. (2017)		
		Oral Cavity	HPV- Oropharynx	HPV+ Oropharynx
T-stage	T0: no primary T1: size ≤2cm T2: size 2-4cm T3: size >4cm T4: ○ T4a: moderately advanced (extrinsic tongue muscle involvement constituted T4a) ○ T4b: very advanced	• T0 deleted • T1: size ≤2cm and DOI ≤5mm • T2: size ≤2cm and DOI 5-10mm or size 2-4cm and DOI ≤10mm • T3: size >4cm or >10mm DOI • T4a extrinsic tongue muscle infiltration now deleted	• T0 deleted	• T0 if proven p16+ disease without evidence of primary tumor • All locally advanced combined to T4
	N0: no LN involved N1: single ipsi LN ≤3cm in size N2: ○ N2a: single ipsi LN, 3-6cm in size ○ N2b: multiple ipsi LNs, all ≤6cm in size ○ N2c: any bi or ctr LNs, all ≤6cm in size N3: any LN >6cm in size	Clinical N-stage		
		• N1-N2 is same as previous and ENE(-) • N3 now with subcategories: ○ N3a is previous N3 (size >6cm) and ENE(-) ○ N3b is any ENE(+), either clinical or radiographic		• Previous N1, N2a combined to N1 (≤6cm with or without ENE) • Previous N2b, N2c combined to N2
		Pathological N-stage		
Stage grouping	Clinical or pathological TNM used for same grouping system	• Microscopically evident ENE(+) LNs results in upstaging		• N1: ≤4 LNs involved • N2: >4 LNs involved • N3 deleted
		Same as previous		Separate clinical and pathological TNM groupings

J. Group Staging for the Larynx, Oropharynx p16(-), Hypopharynx, Oral Cavity, Salivary Glands, Nonmelanoma skin and Paranasal Sinuses

Stage Grouping

Stage 0	Tis	N0	M0
Stage I	T1	N0	M0
Stage II	T2	N0	M0
Stage III	T3	N0	M0
	T1	N1	M0
	T2	N1	M0
	T3	N1	M0
Stage IVA	T4a	N0	M0

TMN Staging for the Oral Cavity/HPV- Continued

	T4a	N1	M0
	T1	N2	M0
	T2	N2	M0
	T3	N2	M0
	T4a	N2	M0
Stage IVB	Any T	N3	M
	T4b	Any N	M0
Stage IVC	Any T	Any N	M1

K. Group Staging for HPV-Mediated (p16+) Oropharyngeal Cancer

Clinical Stage Grouping for HPV-mediated Oropharyngeal Cancer

Stage I	T1	N0, N1	M0
	T2	N0, N1	M0
Stage II	T1	N2	M0
	T2	N2	M0
	T3	N0, N1, N2	M0
Stage III	T4	Any N Stage	M0
	Any T Stage	N3	M0
Stage IV	Any T Stage	Any N Stage	M1

Squamous Cell Carcinoma

- **Prevention**

- Tobacco cessation
- Decrease or eliminate alcohol use
- HPV-vaccine
- Oral screenings
- Education

- **Treatment**

- Surgical excision
 - Chemotherapy
 - Radiation
 - Targeted therapy
 - Cetuximab
 - Immunotherapy
 - Pembrolizumab ([Keytruda](#)) and nivolumab ([Opdivo](#))
- Traditional Methods**

Resources

1. American Cancer Society. “*Targeted Therapy for Oral Cavity and Oropharyngeal Cancer.*” 23, March 2021, <https://www.cancer.org/cancer/types/oral-cavity-and-oropharyngeal-cancer/treating/targeted-therapy.html>
2. American Cancer Society. “*Immunotherapy for Oral Cavity and Oropharyngeal Cancer.*” 23, March 2021, [Immunotherapy for Oral Cavity and Oropharyngeal Cancer | American Cancer Society](#)
3. Cramer, John D et al. “The changing therapeutic landscape of head and neck cancer.” *Nature reviews. Clinical oncology* vol. 16,11 (2019): 669-683.
4. Deschler DG, Moore MG, Smith RV, eds. Quick Reference Guide to TNM Staging of Head and Neck Cancer and Neck Dissection Classification, 4th ed. Alexandria, VA: American Academy of Otolaryngology–Head and Neck Surgery Foundation, 2014.
5. Neville, B. W. (2009). Oral and maxillofacial pathology. St. Louis, Mo: Saunders/Elsevier.
6. Osmosis.
7. Pan C, Rizvi Z. Oral Cancer What the General Surgeon Should Know. *Surg Clin N Am* 102 (2022) 309–324.
8. Yim EK, Park JS. The role of HPV E6 and E7 oncoproteins in HPV-associated cervical carcinogenesis. *Cancer Res Treat.* 2005;37(6):319-324.